

Unit 8 – Heat Transfer in Vessels

- Jacketed Vessels
- Internal Coils
- Agitated Vessels
- Correlations and Calculation Methods

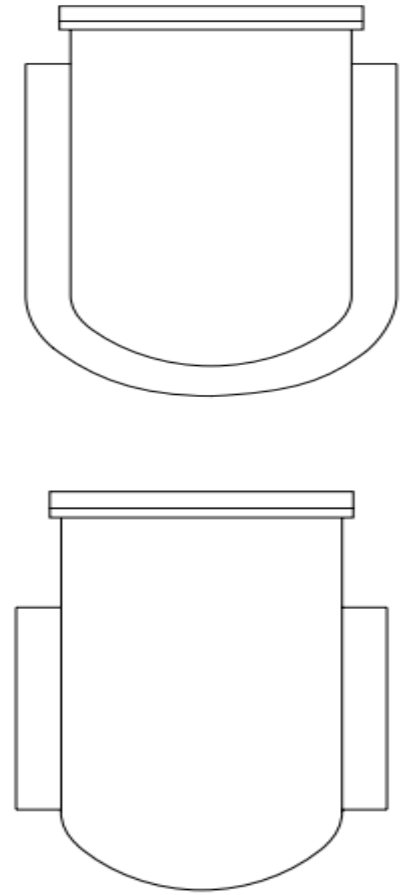
HEAT TRANSFER TO VESSELS

- The simplest way to transfer heat to a process or storage vessel is to fit an external jacket, or an internal coil.

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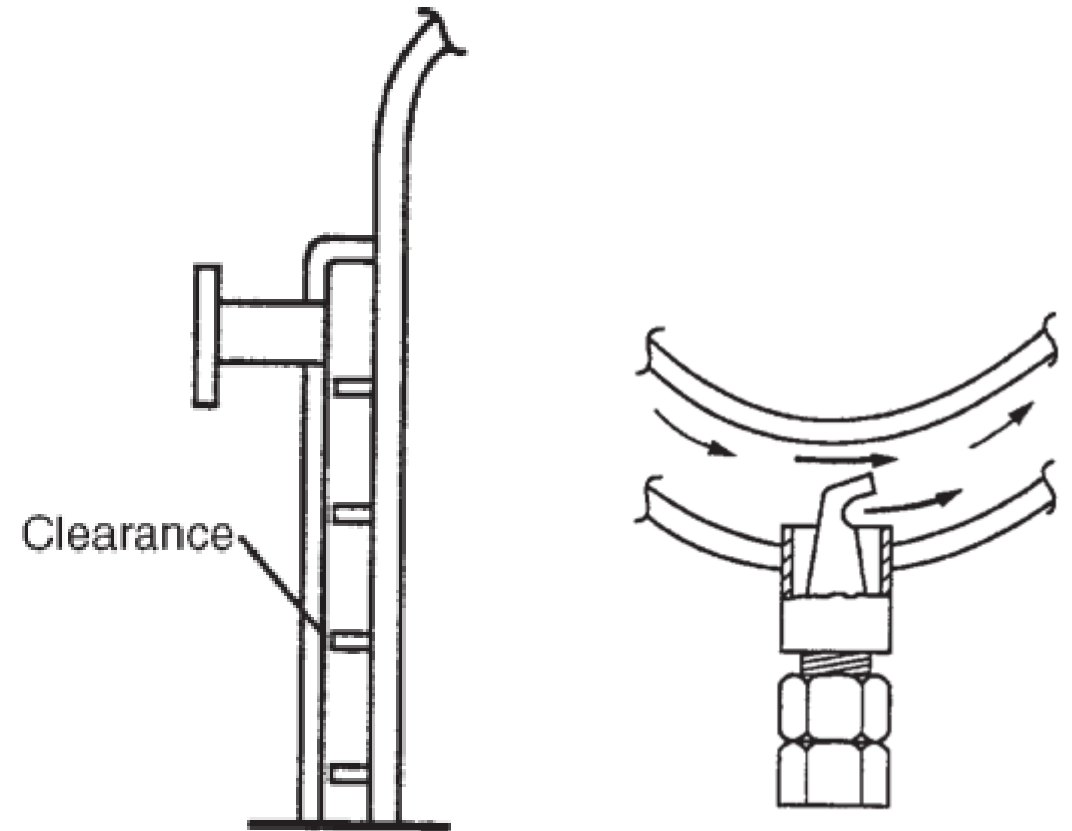
Jacketed vessels - Conventional jackets

- The most commonly used type jacket is that shown in Figure. It consists of an outer cylinder which surrounds part of the vessel.
- The heating or cooling medium circulates in the annular space between the jacket and vessel walls and the heat is transferred through the wall of the vessel.
- Circulation baffles are usually installed in the annular space to increase the velocity of the liquid flowing through the jacket and improve the heat transfer coefficient.



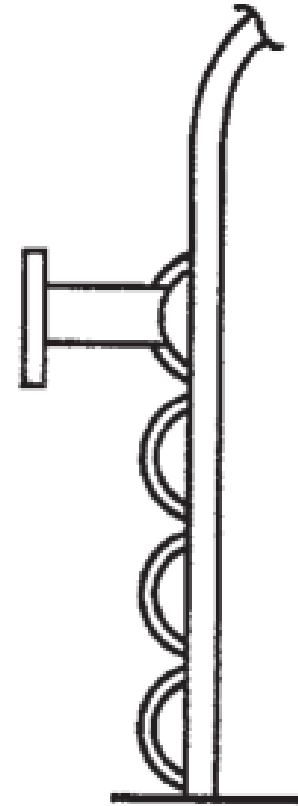
Jacketed vessels - Conventional jackets

- The same effect can be obtained by introducing the fluid through a series of nozzles spaced down the jacket.
- The momentum of the jets issuing from the nozzles sets up a swirling motion in the jacket liquid.
- The spacing between the jacket and vessel wall will depend on the size of the vessel, but will typically range from 50 mm for small vessels to 300 mm for large vessels.



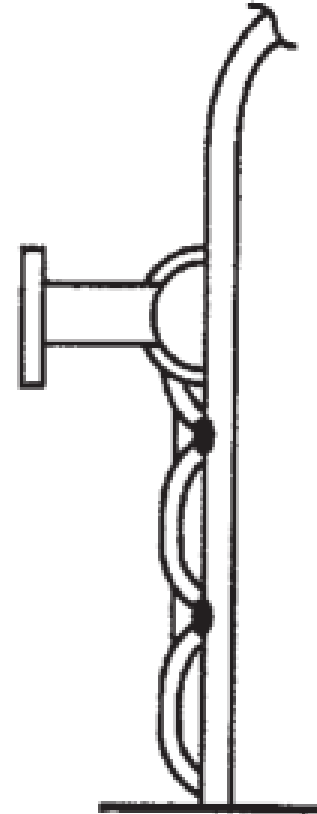
Jacketed vessels - Half-pipe jackets

- Half-pipe jackets are formed by welding sections of pipe, cut in half along the longitudinal axis, to the vessel wall.
- The pipe is usually wound round the vessel in a helix.
- The pitch of the coils and the area covered can be selected to provide the heat transfer area required.
- Standard pipe sizes are used; ranging from 60 to 120 mm outside diameter.
- The half-pipe construction makes a strong jacket capable of withstanding pressure better than the conventional jacket design.



Jacketed vessels - Dimpled jackets

- Dimpled jackets are similar to the conventional jackets but are constructed of thinner plates.
- The jacket is strengthened by a regular pattern of hemispherical dimples pressed into the plate and welded to the vessel wall.



Jacket selection

Factors to consider when selecting the type of jacket to use are listed below:

1. **Cost:** in terms of cost the designs can be ranked, from cheapest to most expensive, as: simple, no baffles agitation nozzles spiral baffle dimple jacket half-pipe jacket
2. **Heat transfer rate required:** select a spirally baffled or half-pipe jacket if high rates are required.
3. **Pressure:** as a rough guide, the pressure rating of the designs can be taken as: jackets, up to 10 bar dimpled jackets, up to 20 bar half-pipe, up to 70 bar. So, half-pipe jackets would be used for high pressure.

Jacket heat transfer and pressure drop

- The heat transfer coefficient to the vessel wall can be estimated using the correlations for forced convection in conduits.

$$Nu = CRe^{0.8}Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

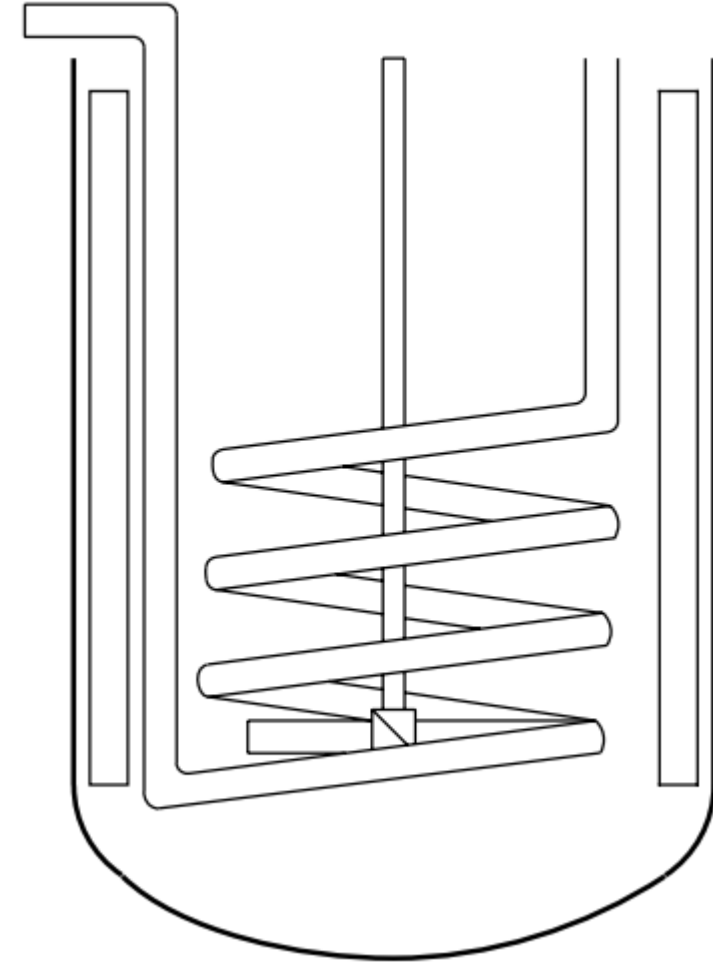
- Where $C = 0.021$ for gases, $= 0.023$ for non-viscous liquids, $= 0.027$ for viscous liquids
- The fluid velocity and the path length can be calculated from the geometry of the jacket arrangement.
- The hydraulic mean diameter (equivalent diameter, d_e) of the channel or half-pipe should be used as the characteristic dimension in the Reynolds and Nusselt numbers.

Jacket heat transfer and pressure drop

- In dimpled jackets a velocity of 0.6 m can be used to estimate the heat transfer coefficient.
- A method for calculating the heat transfer coefficient for dimpled jackets is given by Makovitz (1971).
- The coefficients for jackets using agitation nozzles will be similar to that given by using baffles.
- A method for calculating the heat transfer coefficient using agitation nozzles is given by Bolliger (1982).
- To increase heat transfer rates, the velocity through a jacket can be increased by recirculating the cooling or heating liquid.
- For simple jackets without baffles, heat transfer will be mainly by natural convection and the heat transfer coefficient will range from 200 to 400 W/m²°C.

Internal coils

- The simplest and cheapest form of heat transfer surface for installation inside a vessel is a helical coil.
- The pitch and diameter of the coil can be made to suit the application and the area required.
- The diameter of the pipe used for the coil is typically equal to $D_v/30$, where D_v is the vessel diameter.
- The coil pitch is usually around twice the pipe diameter.
- Small coils can be self supporting, but for large coils some form of supporting structure will be necessary.
- Single or multiple turn coils are used.



Coil heat transfer and pressure drop

- The heat transfer coefficient at the inside wall and pressure drop through the coil can be estimated using the correlations for flow through pipes.
- Correlations for forced convection in coiled pipes are also given in the Engineering Sciences Data Unit Design Guide, ESDU 78031 (2001).

Overall Heat Transfer Coefficient

- For coils and jackets, the Overall Heat Transfer Coefficient can be calculated

$$\frac{1}{U} = \frac{1}{\alpha} + \frac{\delta}{\lambda} + \frac{1}{\alpha_s} + \frac{1}{\alpha_f}$$

- where α and α_s are the process and service side heat transfer coefficients, respectively.
- The service side fouling resistance, $1/\alpha_f$, will be available from local experience or from Kern (1950).
- A typical overall coefficient for a well designed coil would be 400 to 600 W/m²K.

Overall Heat Transfer Coefficient

- **Typical overall coefficients for jacketed glass lined steel vessels**

Duty	U ($\text{W m}^{-2}\text{K}^{-1}$)
Distillation/Evaporation	350
Heating	310
Cooling	200
Cooling (chilled service)	100

Overall Heat Transfer Coefficient

- Typical overall coefficients for jacketed carbon and stainless steel vessels

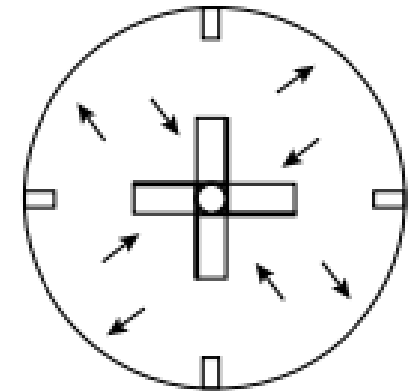
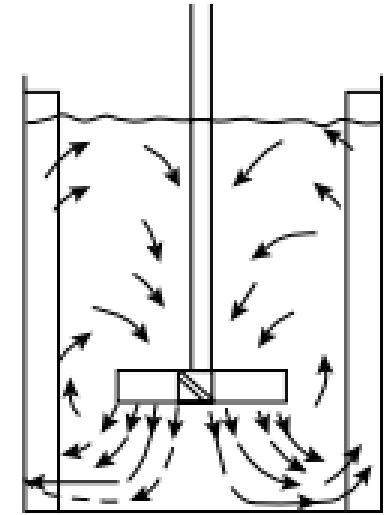
Duty	U ($\text{W m}^{-2}\text{K}^{-1}$)
Heating	400
Cooling	350
Cooling (chilled service)	150

Agitated vessels

- Unless only small rates of heat transfer are required, as when maintaining the temperature of liquids in storage vessels, some form of agitation will be needed.
- The various types of agitator used for mixing and blending are also used to promote heat transfer in vessels.

Stirred tanks

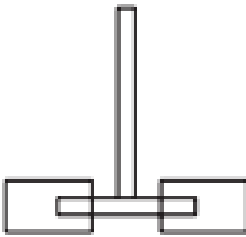
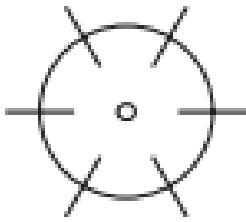
- Mixing vessels fitted with some form of agitator are the most commonly used type of equipment for blending liquids and preparing solutions.
- A typical arrangement of the agitator and baffles in a stirred tank, and the flow pattern generated.
- Mixing occurs through the bulk flow of the liquid and, on a microscopic scale, by the motion of the turbulent eddies created by the agitator.
- Bulk flow is the predominant mixing mechanism required for the blending of miscible liquids and for solids suspension.
- Turbulent mixing is important in operations involving mass and heat transfer; which can be considered as shear-controlled processes.



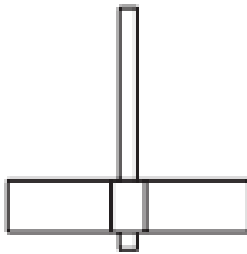
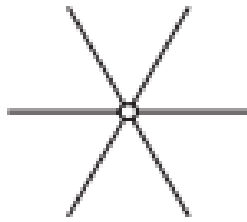
Stirred tanks

- The most suitable agitator for a particular application will depend on the type of mixing required, the capacity of the vessel, and the fluid properties, mainly the viscosity.
- The three basic types of impeller which are used at high Reynolds numbers (low viscosity).
- They can be classified according to the predominant direction of flow leaving the impeller.
- The flat-bladed (Rushton) turbines are essentially radial-flow devices, suitable for processes controlled by turbulent mixing (shear-controlled processes).
- The propeller and pitched-bladed turbines are essentially axial-flow devices, suitable for bulk fluid mixing.
- Paddle, anchor and helical ribbon agitators and other special shapes, are used for more viscous fluids.

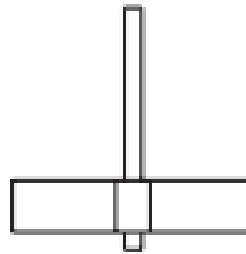
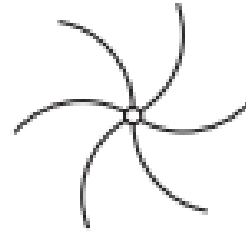
Stirred tanks - Turbine impeller



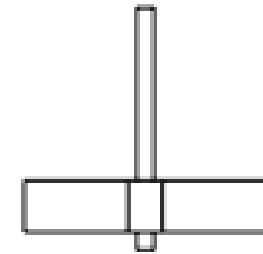
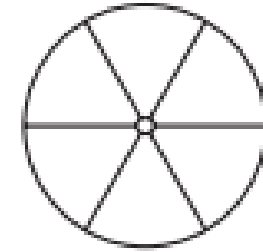
Disc-mounted flat-blade turbine



Hub-mounted flat-blade turbine

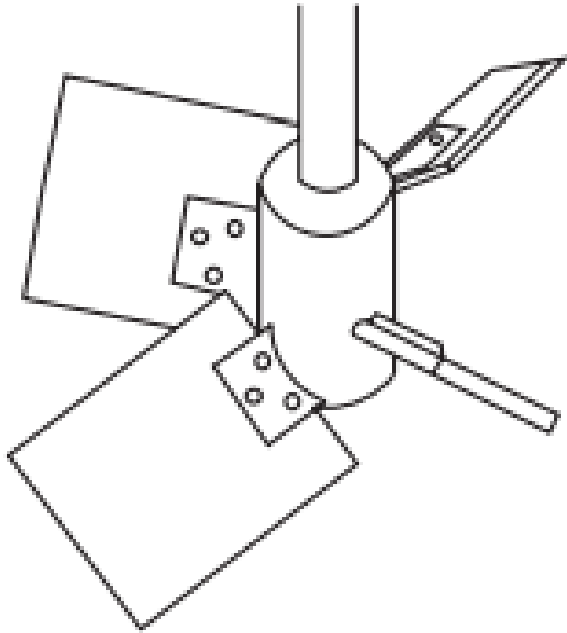


Hub-mounted curved-blade turbine

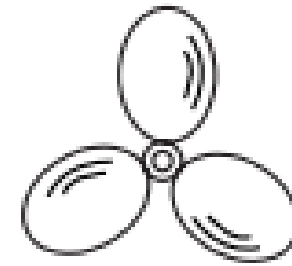


Shrouded turbine impeller

Stirred tanks - (b) Pitched bladed turbine (c) Marine propeller

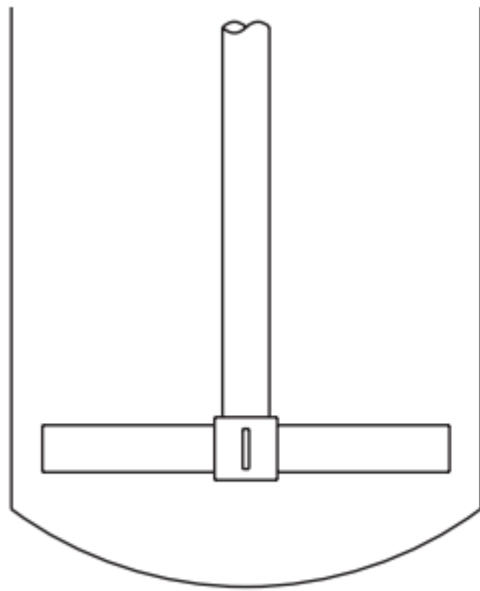


(b)

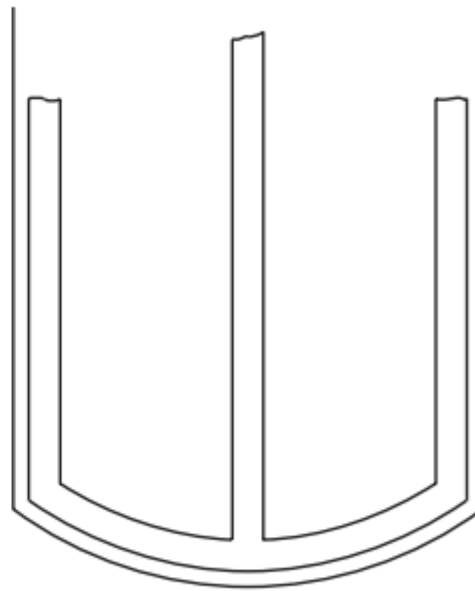


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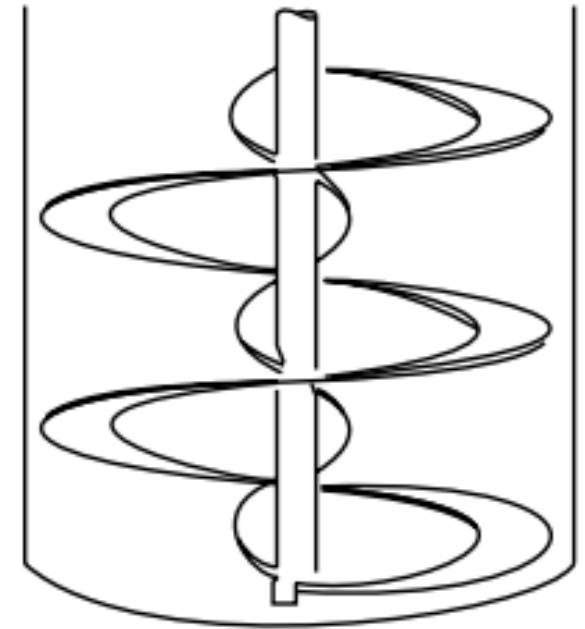
Stirred tanks - Low-speed agitators (a) Paddle (b) Anchor (c) Helical ribbon



(a)



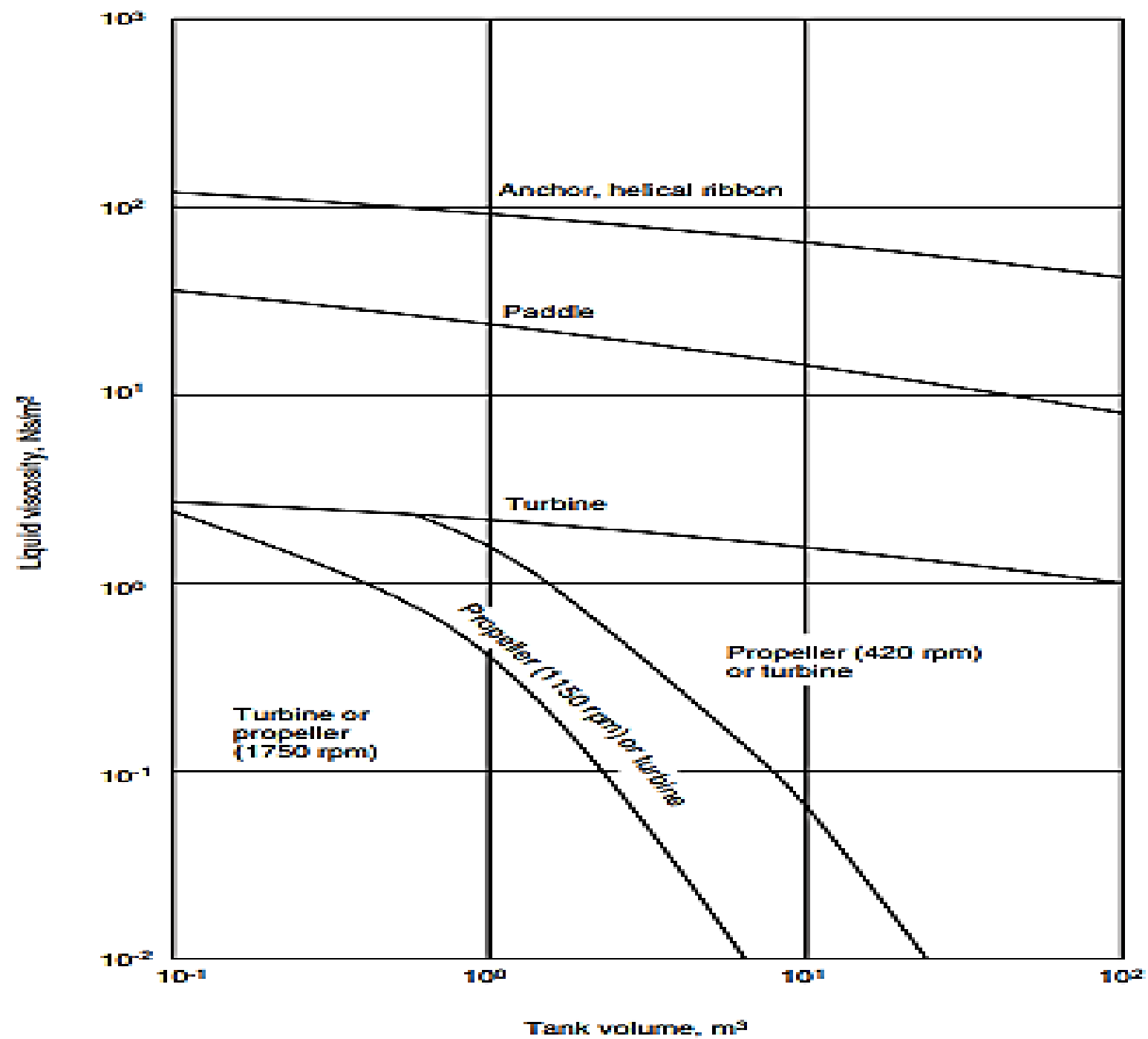
(b)



(c)

Agitator selection

- The selection chart which has been adapted from a similar chart given by Penney (1970), can be used to make a preliminary selection of the agitator type, based on the liquid viscosity and tank volume.
- For turbine agitators, impeller to tank diameter ratios of up to about 0.6 are used, with the depth of liquid equal to the tank diameter.
- Baffles are normally used, to improve the mixing and reduce problems from vortex formation.
- Anchor agitators are used with close clearance between the blades and vessel wall, anchor to tank diameter ratios of 0.95 or higher.



Correlations and Calculation Methods

- The correlations used to estimate the heat transfer coefficient to the vessel wall, or to the surface of coils, have the same form as those used for forced convection in conduits.
- The fluid velocity is replaced by a function of the agitator diameter and rotational speed, $D \times N$, and the characteristic dimension is the agitator diameter.

$$Nu = CRe^a Pr^b \left(\frac{\mu}{\mu_w} \right)^c$$

Correlations and Calculation Methods

- For agitated vessels:

$$\frac{h_v D}{k_f} = C \left(\frac{N D^2 \rho}{\mu} \right)^a \left(\frac{C_p \mu}{k_f} \right)^b \left(\frac{\mu}{\mu_w} \right)^c$$

where h_v = heat transfer coefficient to vessel wall or coil, $\text{Wm}^{-2}\text{°C}^{-1}$

D = agitator diameter, m

N = agitator, speed, rps (revolutions per second)

ρ = liquid density, kg/m^3

k_f = liquid thermal conductivity, $\text{Wm}^{-1}\text{°C}^{-1}$

C_p = liquid specific heat capacity, $\text{J kg}^{-1}\text{°C}^{-1}$

μ = liquid viscosity, Nm^{-2}s .

The values of constant C and the indices a , b and c depend on the type of agitator, the use of baffles, and whether the transfer is to the vessel wall or to coils. Baffles will normally be used in most applications.

Correlations and Calculation Methods

- Flat blade paddle, baffled or unbaffled vessel, transfer to vessel wall, $Re < 4000$:

$$Nu = 0.36Re^{0.67}Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

- Flat blade disc turbine, baffled or unbaffled vessel, transfer to vessel wall, $Re < 400$:

$$Nu = 0.54Re^{0.67}Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

- Flat blade disc turbine, baffled vessel, transfer to vessel wall, $Re > 400$:

$$Nu = 0.74Re^{0.67}Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

Correlations and Calculation Methods

- Propeller, 3 blades, transfer to vessel wall, $Re > 5000$:

$$Nu = 0.64 Re^{0.67} Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

- Turbine, flat blades, transfer to coil, baffled, $Re = 2000 - 700,000$:

$$Nu = 1.10 Re^{0.62} Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

- Paddle, flat blades, transfer to coil, baffled:

$$Nu = 0.87 Re^{0.62} Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

- More comprehensive design data is given by: Uhl and Gray (1967), Wilkinson and Edwards (1972), Penny (1983) and Fletcher (1987).

Question: Find the heat transfer coefficient for the agitated vessel with propeller: for mineral oil density = 860 kg/m³, Cp = 2000 J/kg.K, thermal conductivity k = 0.14 W/mK, viscosity = 20 mPa.s, diameter of propeller = 70 cm, speed = 10 cm/s. Viscosity of water = 1 mPa.s

$$Nu = 0.64 Re^{0.67} Pr^{0.33} \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

$$Re = \frac{\rho v d}{\mu} = \frac{860 \times 10 / 100 \times 70 / 100}{20 \times 10^{-3}} = 3010$$

$$Pr = \frac{\mu C_p}{k_f} = \frac{20 \times 10^{-3} \times 2000}{0.14} = 285.7$$

$$Nu = 0.64 \times 3010^{0.67} \times 285.7^{0.33} \times \left(\frac{20 \times 10^{-3}}{1 \times 10^{-3}} \right)^{0.14} = 1347.1$$

Question: Find the heat transfer coefficient for the agitated vessel with propeller: for mineral oil density = 860 kg/m³, Cp = 2000 J/kg.K, thermal conductivity k = 0.14 W/mK, viscosity = 20 mPa.s, diameter of propeller = 70 cm, speed = 10 cm/s. Viscosity of water = 1 mPa.s

$$Nu = \frac{hD}{k_f}$$

$$h = \frac{Nu \times k_f}{D} = \frac{1347.1 \times 0.14}{70/100} = 269.4 \text{ W/m}^2\text{K}$$